Introduction to Computer Science and Programming

Labo 8 – VS CODE and C++ Exercises

Laboratory 8, December 2, 2025

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Today…

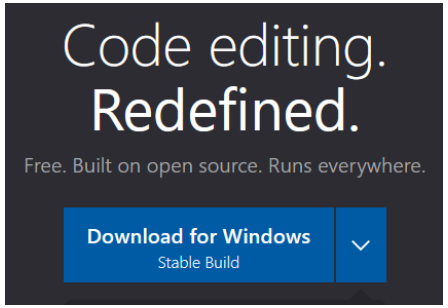
Today's lab is dedicated to **VS CODE and first examples/exercises in C++**

Covered topics in the today evaluated lab:

- Quick intro to VS CODE
  - Professional IDE!
    - We will use a subset of the features
- Setting up VS CODE
- C/C++ for VS CODE
- C++ exercises

Visual Studio Code

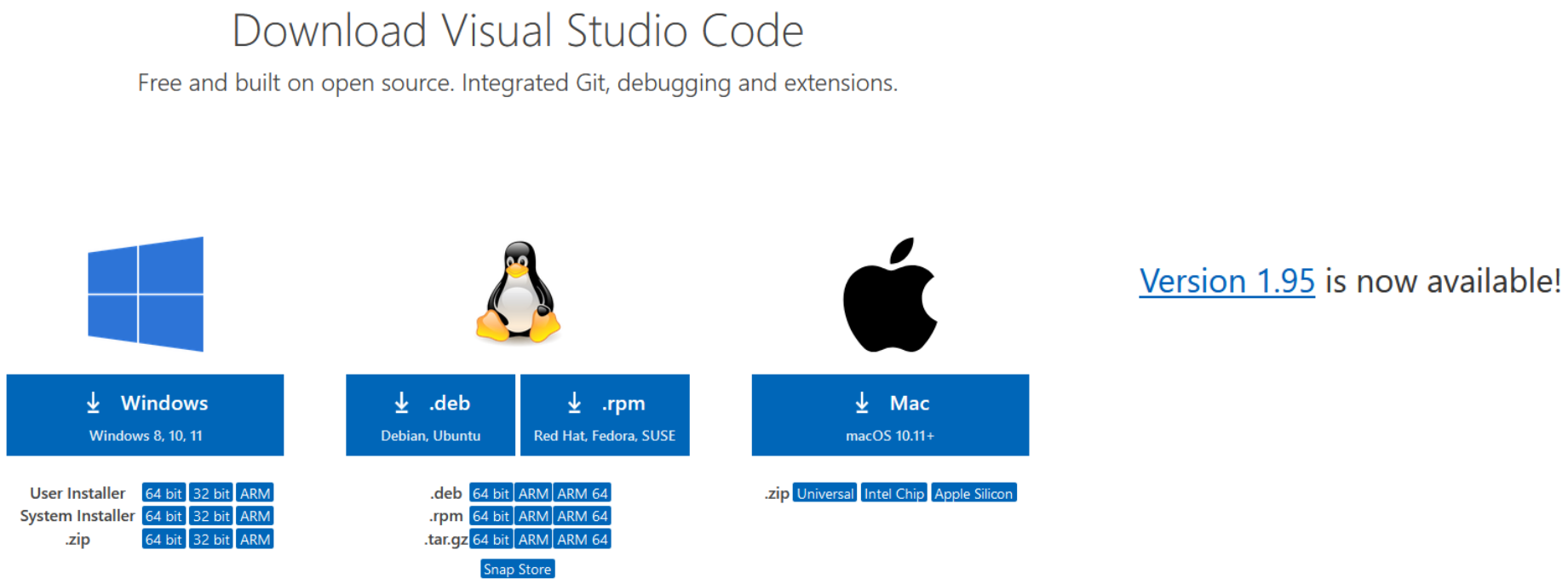
What is VS Code?



- Visual Studio Code is a **cross-platform and multilanguage source code editor** developed by Microsoft
  - **Microsoft, Linux, macOS**
  - **C++, Java, Python, Javascript, …**
- It incorporates the capabilities of Visual Studio while making them **lightweight**
- Launched in 2015, Visual Studio Code marked Microsoft's move toward the **open-source** community
- The source code is available at this link: https://github.com/microsoft/vscode

Cross-platform source code editor

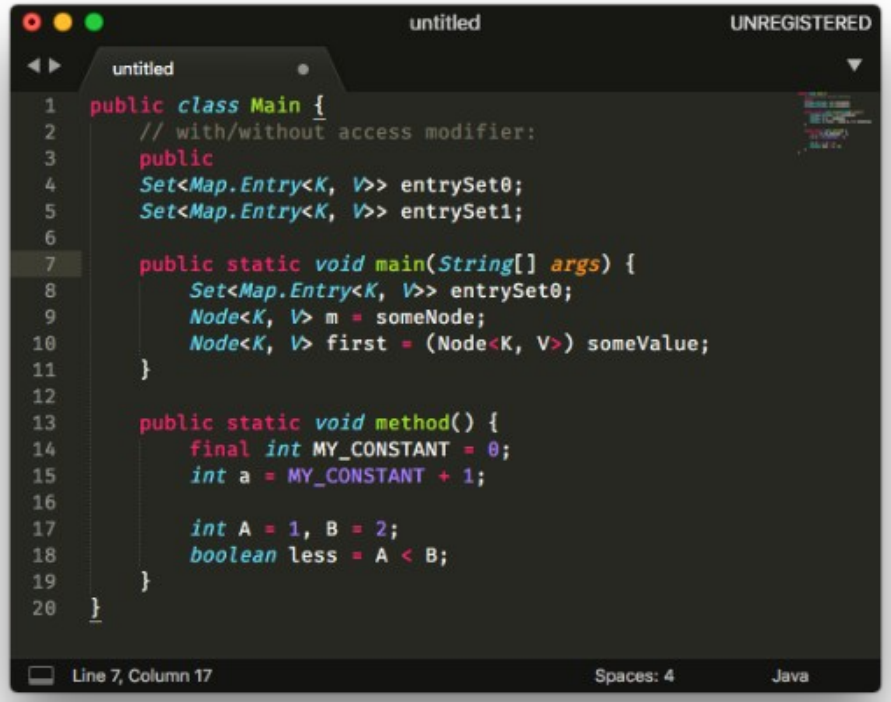
- Visual Studio Code is natively supported on **Linux**, **Windows**, and **macOS** environments



https://code.visualstudio.com/Download

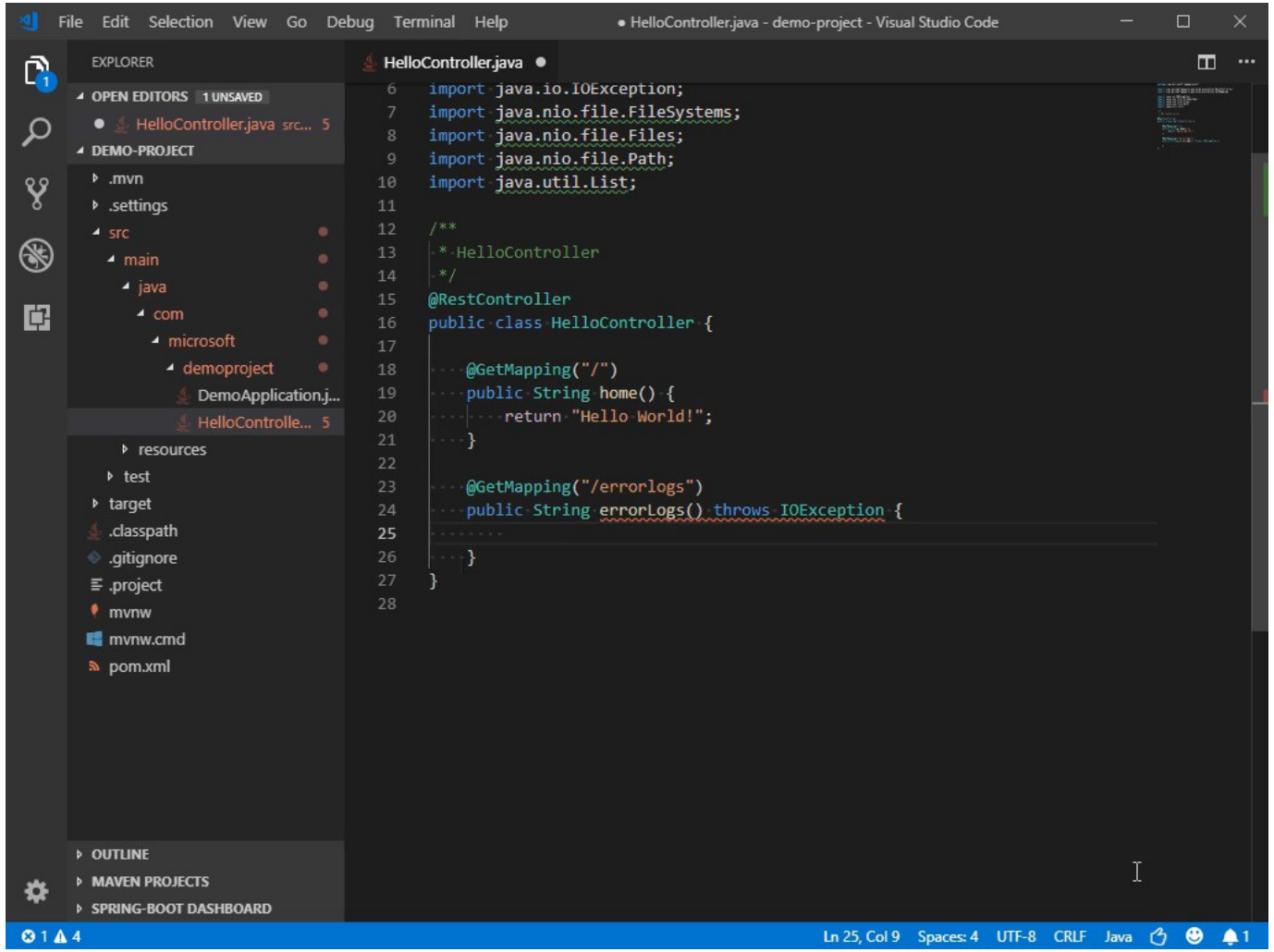
What are its most interesting features?

- **Syntax Highlight**: Syntax coloring
- **IntelliSense**: Autocompletion with documentation
- **Integrated GIT**: Built-in version control software
- **Integrated Debugging**: Error detection and correction with breakpoint definition
- **Extensions (plug-ins):** Ability to easily and quickly add additional features



We will understand these features better in the upcoming lessons

IntelliSense (Code Completion)



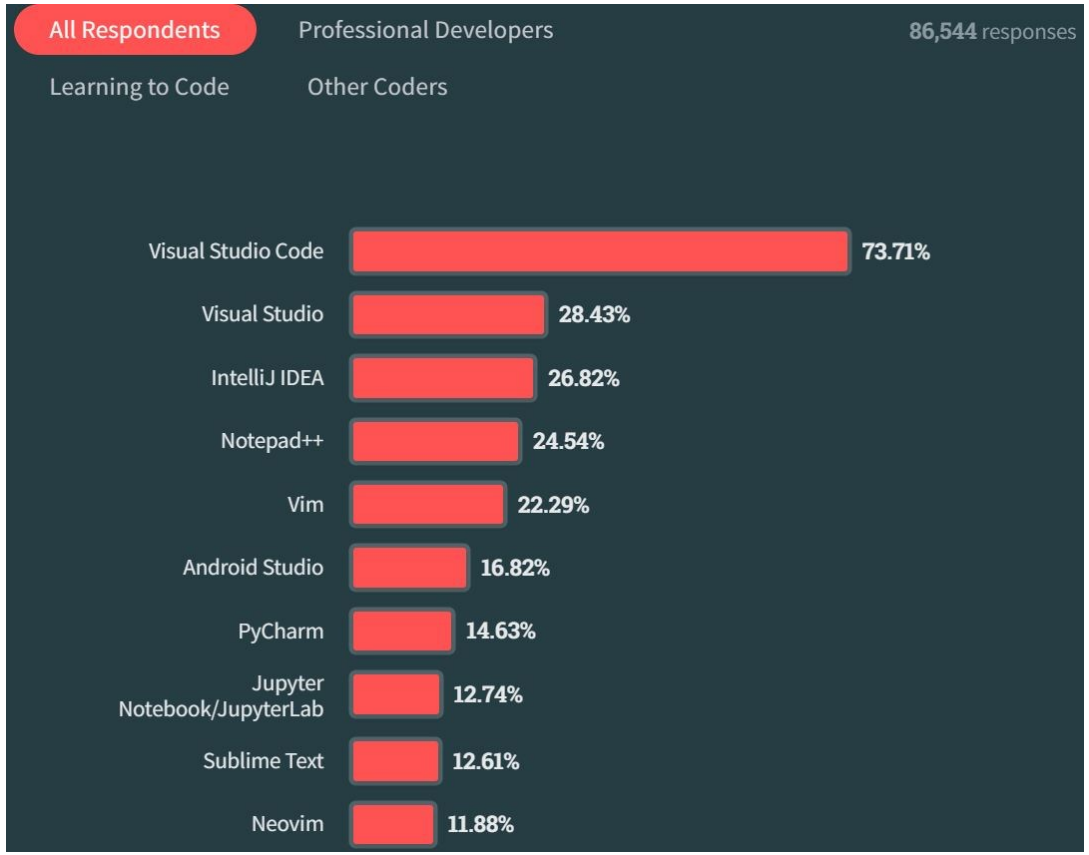
It helps speed up the coding phase

Why was VS Code chosen?

- **Cross-Platform Compatibility**: Works seamlessly on **Windows**, **macOS**, and **Linux**, providing a consistent experience across different systems
- **IntelliSense and Autocompletion**: Offers **smart code suggestions**, syntax highlighting, and inline documentation, boosting productivity and reducing errors
- **Integrated Version Control**: Built-in Git support allows for easy version control, commit management, and **collaboration directly within the editor**
- **Extensive Extension Marketplace**: A vast **library of extensions and plug-ins** enables customization and additional functionality for different languages and tools
- **Lightweight and Fast**: Despite its rich features, VS Code is **lightweight and responsive**, making it ideal for both large and small projects
- **It is widely used in practice in companies!**

Stack Overflow 2023 Developer Survey

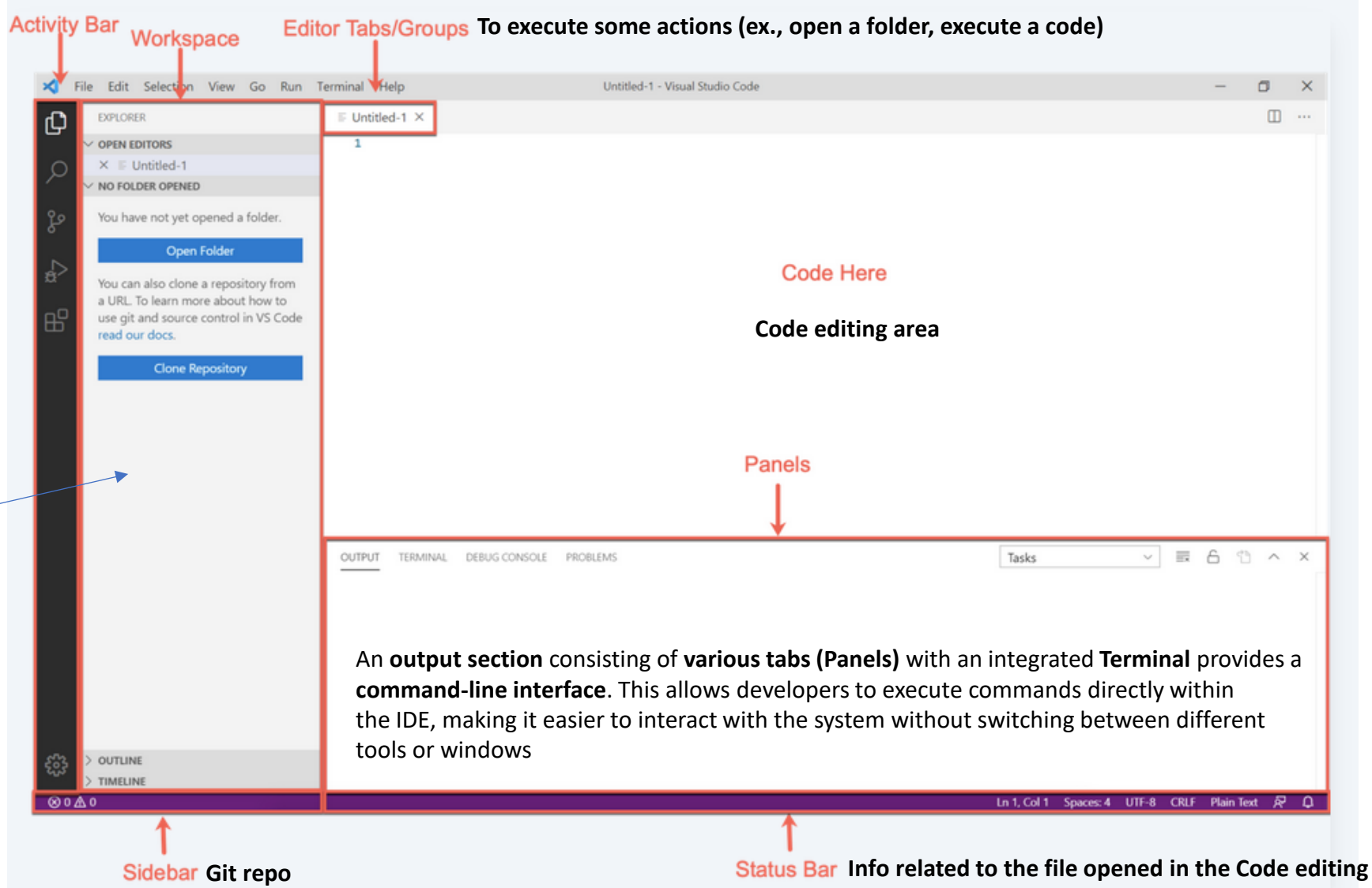
- Visual Studio Code remains the **preferred IDE across all developers**, increasing its use among all developers this year from 75 percent to 81 percent



VS CODE GUI

The **main navigation area** in **VS Code**

The content here depends on the activity For example with Explorer here we have Folder and Files of the project (and other views)

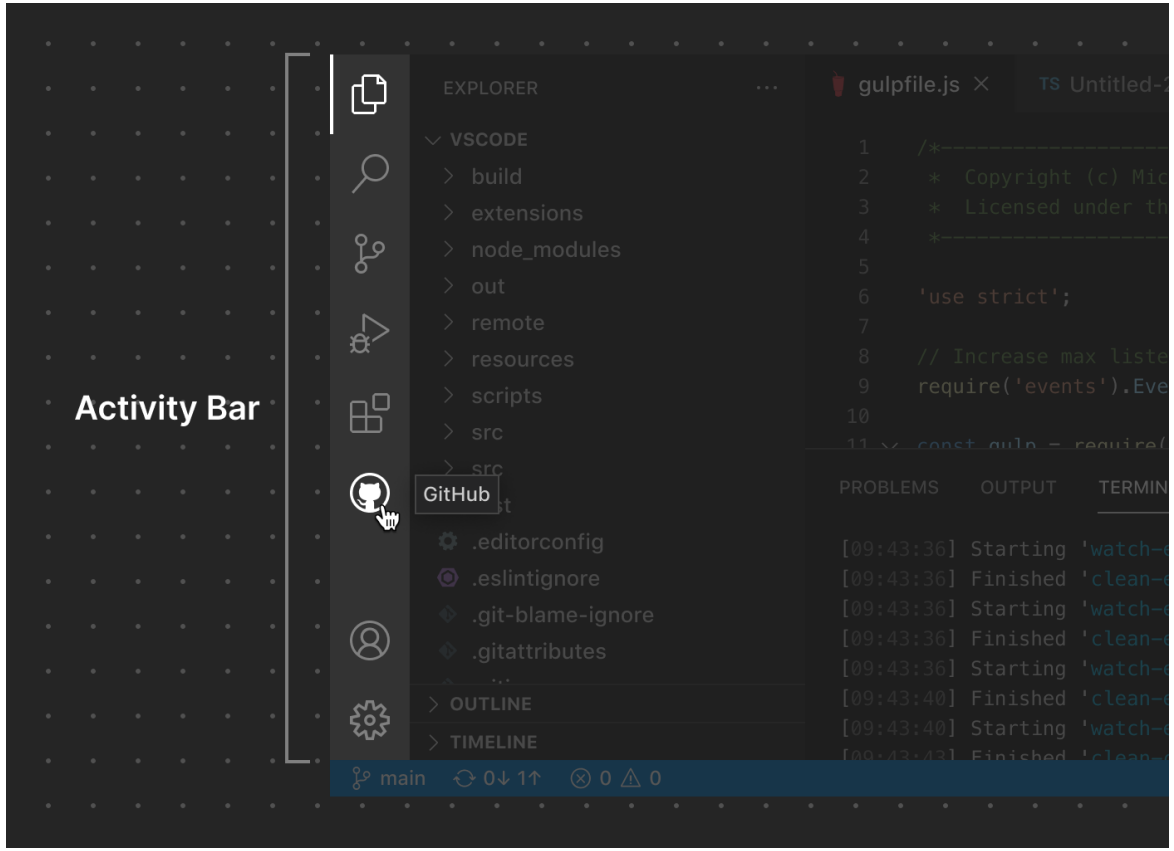


Activity Bar

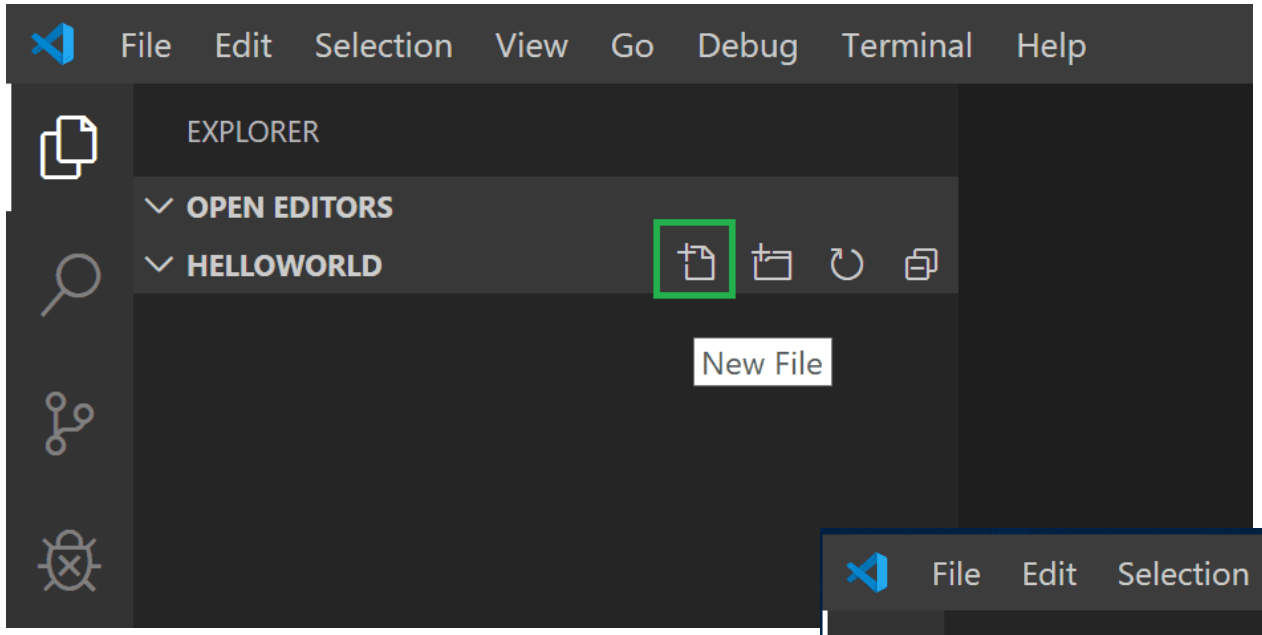
- The **Activity Bar** is the main navigation area in **VS Code**, located on the side of the interface. It allows quick access to core features like the **Explorer**, **Search**, **Source Control**, and **Extensions**

Explorer view
Search
Source Control (Git)
Run and Debug
Extensions
GitHub

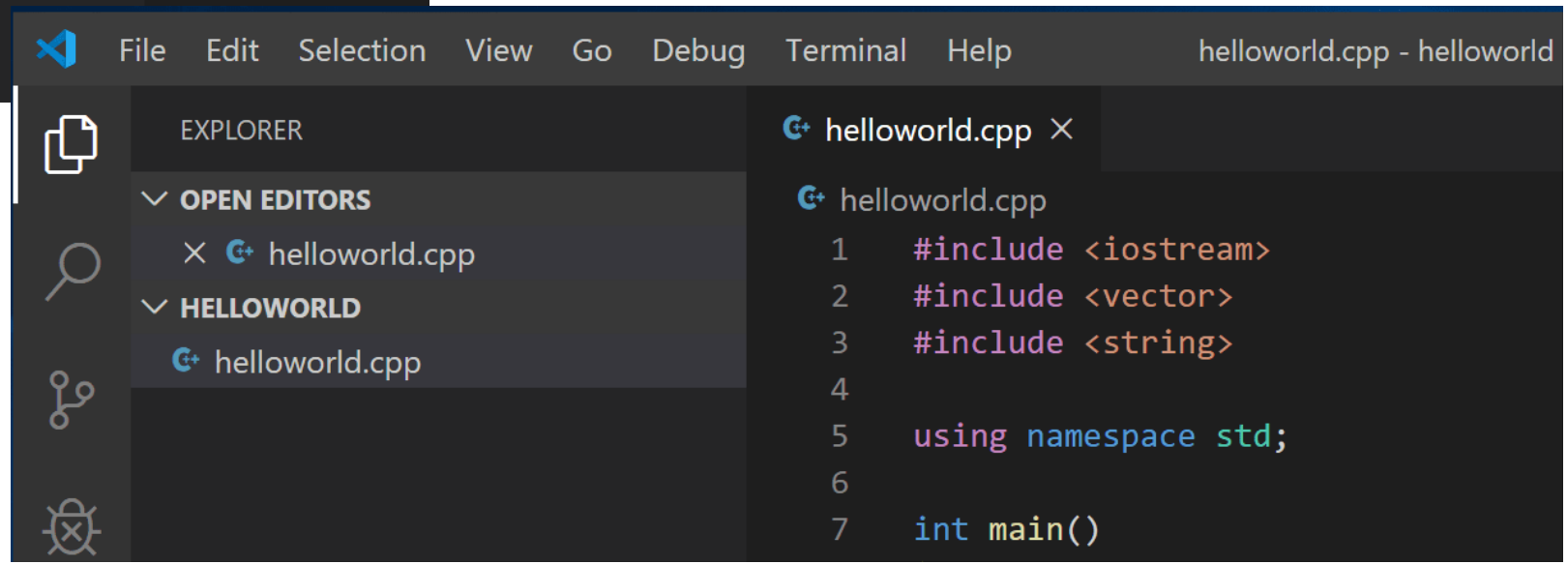
Feature added as extension



Explorer View

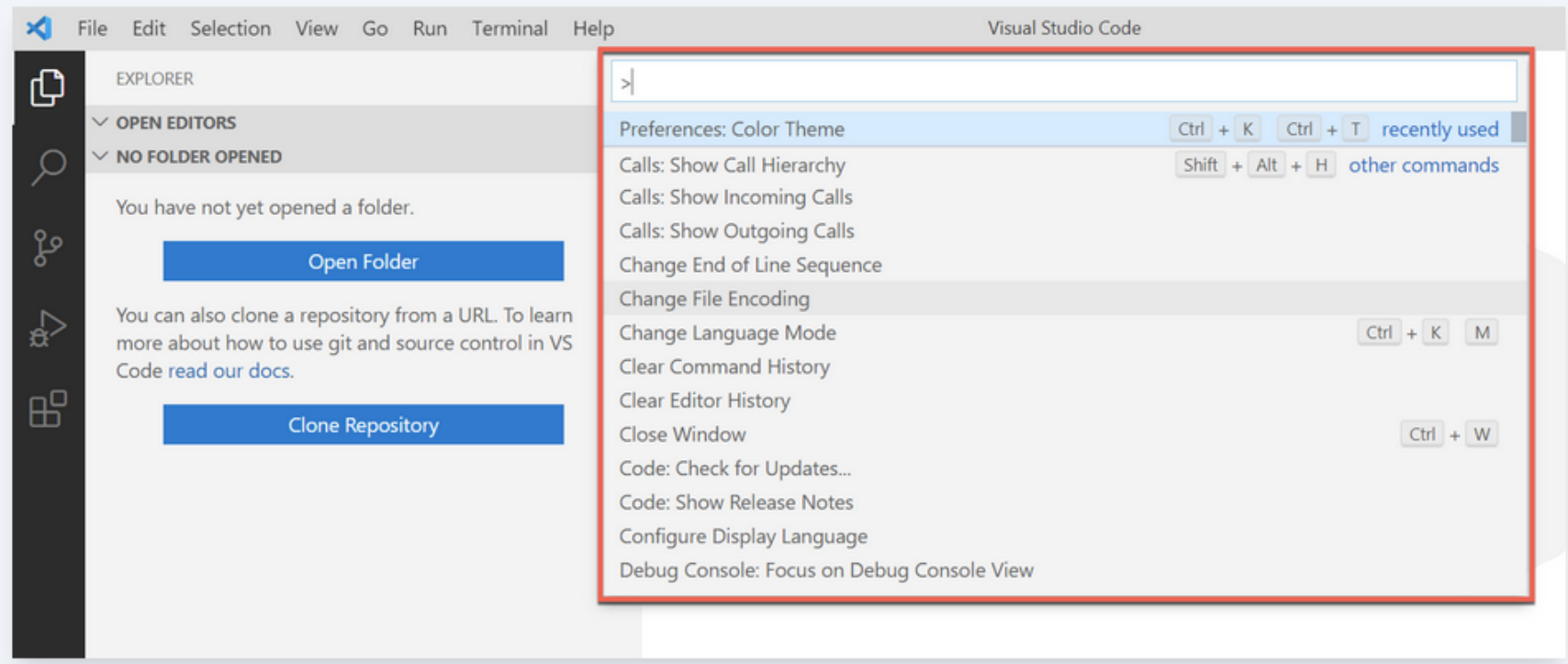


The Explorer view is *used to browse, open, and manage the files and folders in your project*

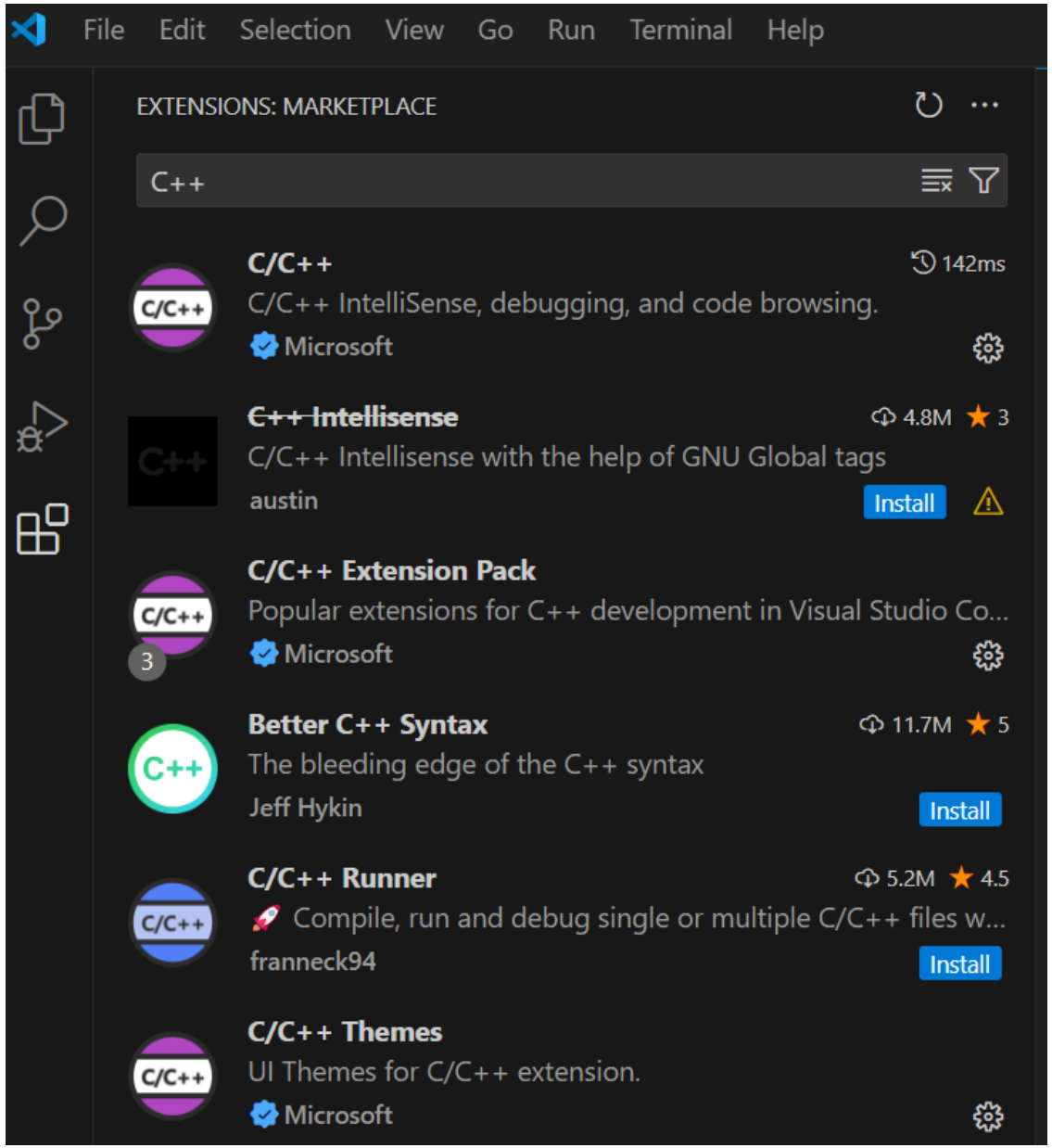


Command Palette

- **Ctrl+Shift+P**
- From here, you have access to all functionality within VS Code, including keyboard shortcuts for the most common operations

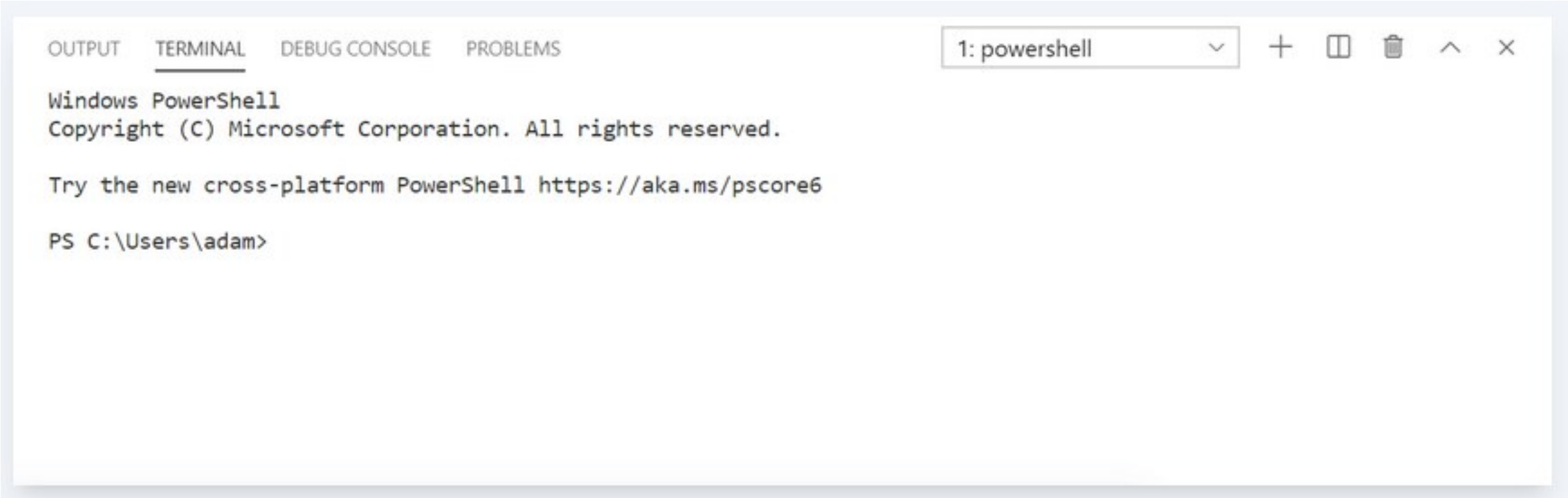


Extensions


File Edit Selection View Go Run Terminal Help
EXTENSIONS: MARKETPLACE
C++
C/C++
142ms
C/C++ IntelliSense, debugging, and code browsing.
Microsoft
C++ Intellisense
4.8M 3
C/C++ Intellisense with the help of GNU Global tags
austin
Install
C/C++ Extension Pack
Popular extensions for C++ development in Visual Studio Co...
Microsoft
3
Better C++ Syntax
11.7M 5
The bleeding edge of the C++ syntax
Jeff Hykin
Install
C/C++ Runner
5.2M 4.5
Compile, run and debug single or multiple C/C++ files w...
franneck94
Install
C/C++ Themes
UI Themes for C/C++ extension.
Microsoft

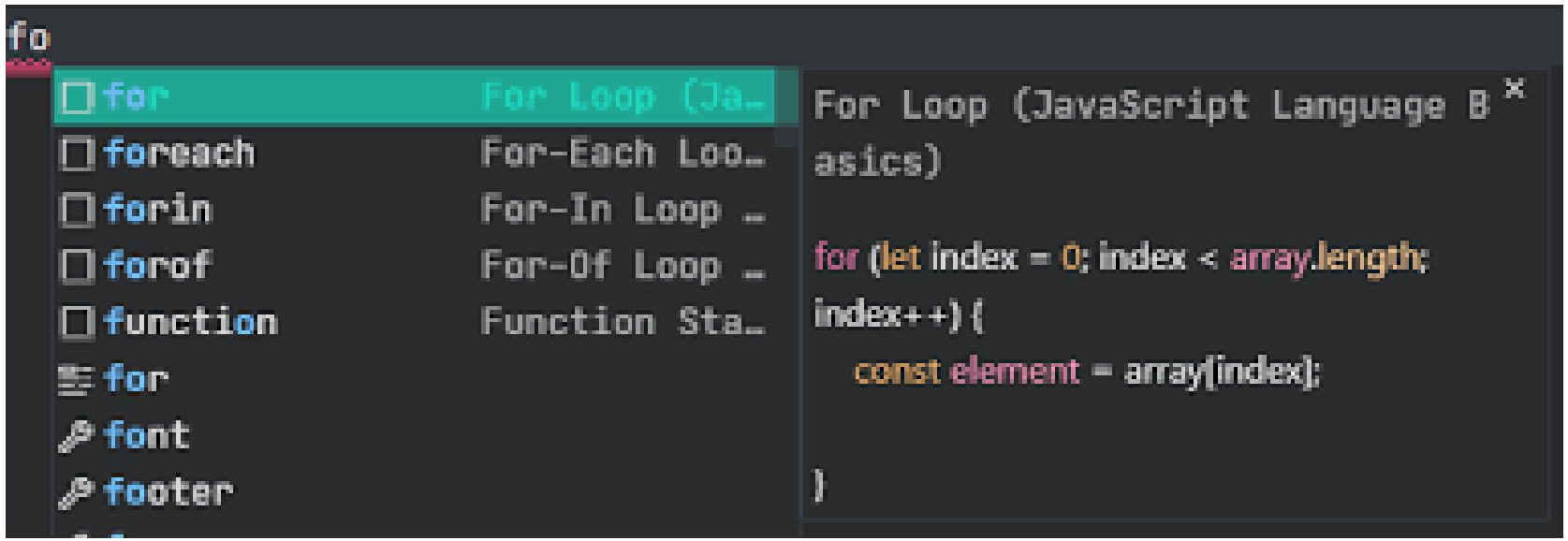
Terminal (CLI)

- You can use the integrated terminal just like a typical console by typing commands, obtaining output, and performing various tasks directly within the editor



Snippets

- Code snippets are **templates** that make it easier to enter repeating code patterns, such as loops or conditional-statements

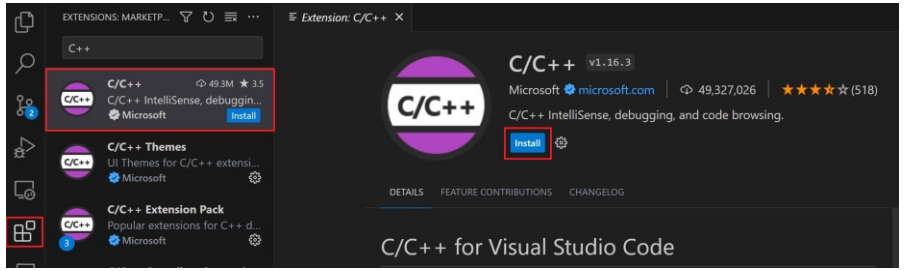


Setting up Visual Studio Code

- Go to https://code.visualstudio.com/docs/setup/setup-overview
- Select the Operating system
- Follow the instructions!
- Suggestions:
  - Select all default settings
  - Choose the look & feel of the GUI

How many of you have not yet installed VS Code?

C/C++ for Visual Studio Code



- Install the C++ extension in VS CODE
- The C/C++ extension doesn't include a C++ compiler or debugger
- Install a Compiler
  - Different options
  - **For Windows: MingGV**

Windows:

Go to the MSVC tutorial

Go to the MinGW tutorial

Linux:

Go to the GCC tutorial

macOS:

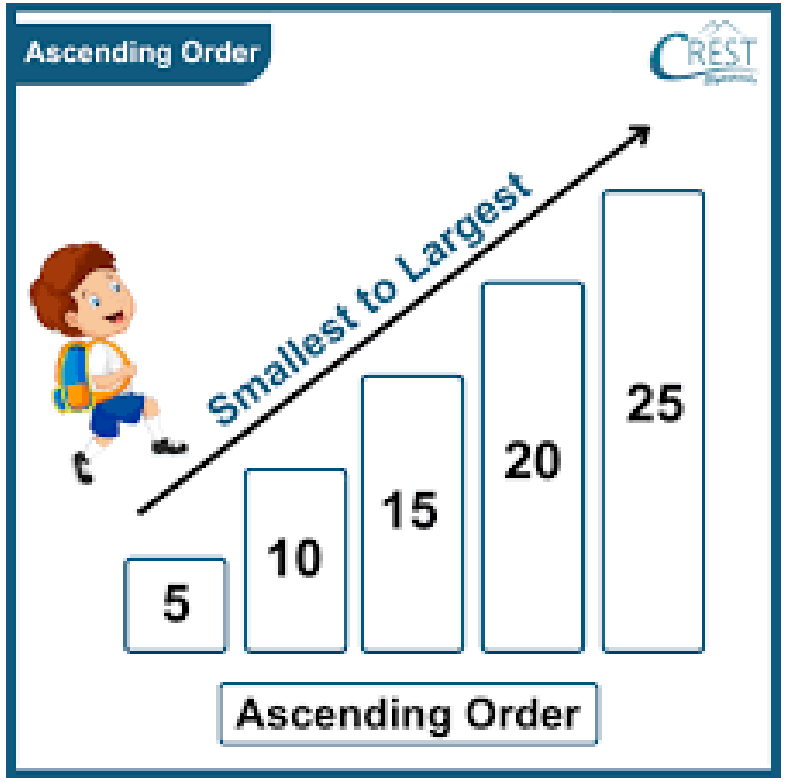
Go to the Clang tutorial

How many of you have not yet installed a Compiler?
Problems during installation?

https://code.visualstudio.com/docs/languages/cpp

Exercise 1: PRINT THREE NUMBERS IN ASCEDNING ORDER

- Write a C++ program that reads **five integers** and prints them in **ascending order**
- For example:
  - Input: 10, 20, 15, 5, 25
  - Output: 5, 10, 15, 20, 25



Exercise 1: PRINT THREE NUMBERS IN ASCEDNING ORDER

- Following is the proposed algorithm given in pseudocode (which specifies additional details).

- For example:

Output: 5, 10, 15, 20, 25

```
// Prompt the user to enter three integers
// Declare three int variables: a0, a1, and a2
// Read the values of a0, a1, and a2
// Sort a0, a1, and a2 as follows:
// Declare an integer variable aux initialized with a1
// If a0 is greater than a1, swap a0 and a1, meaning:
// - Assign a1 the value of a0, a0 the value of aux, and...
// ... aux the value of a1 (at this point, a0 <= a1 == aux)
// If a0 is greater than a2:
// - Assign a1 the value of a0, a0 the value of a2, and...
// ... a2 the value of aux
// Otherwise:
// - If a1 is greater than a2, swap a1 and a2, meaning:
// -- Assign a1 the value of a2, and a2 the value of aux...
// ... (at this point, a0 <= a1 <= a2)
// Print the string "The entered numbers in ascending order are: "
// Print the value of a0, followed by the character <
// Print the value of a1, followed by the character <
// Print the value of a2
// Print a newline
```

Exercise 2: DISPLAYING TIME

- The time(0) function in the ctime header file returns the current time in seconds elapsed since the time 00:00:00 on January 1, 1970 GMT

- Write a C++ program that displays the current time in the format hour:minute:second

Exercise 3: FARE CALCULATION

- Write a C++ program that reads the number of kilometers traveled by a taxi.

- Example:
  Input: 22
  Output: Final price based on the rules in euros.

The fare rules are:

Base fare: 5 €

Each kilometer costs:

2 € if the distance is 10 km or less

1.50 € if the distance is more than 10 km

Exercise 4: AVERAGE GRADE CALCULATION

- Write a C++ program that reads three test scores and calculates the average grade.

- **Grading Scale:**

  A: 90-100

  B: 80-89

  C: 70-79

  D: 60-69

  F: Below 60

Requirements:

- Calculate the average as a decimal number
- Display both the average and the letter grade
- Add a message like "Excellent!" for A, "Good job!" for B, etc.

Exercise 5: CELSIUS TO FAHRENHEIT CONVERSION

- Write a C++ program that converts a temperature from Celsius to Fahrenheit.

Formula:

$$F = (C \times \frac{9}{5}) + 32$$

Example:

- 0°C should give 32°F (freezing point)
- 100°C should give 212°F (boiling point)
- 37°C should give 98.6°F (body temperature)
- -40°C should give -40°F (interesting coincidence!)